

Inference about means

Slide route: Lecture2v2.tex: hypothesis testing through California Test Score Data Set. Stock and Watson Chapter 3.

Session 3 | Evidence against a claim

Outcomes: State a null and alternative, compute a test statistic and confidence interval, distinguish statistical from economic significance, and compare two independent groups.

Opening A sample mean is 104; a policy report claims the population mean is 100. Is four units a large discrepancy? We need the scale of sampling uncertainty.

100-minute route

0–10	Recall sampling distributions and SE.
10–30	Hypothesis Testing, Key Concepts, Computing the P-Value.
30–45	Sample variance, standard errors, t-statistic; worked test.
45–55	Type I/II errors and short pause.
55–70	Confidence intervals and test/interval correspondence.
70–90	Mean Difference Testing, randomization, California application.
90–100	Exit exercise and preview of a regression slope.

Slide-by-slide script

Slide 1 | Title

Guide Open with the question: how do we decide whether a population claim is compatible with sample evidence?

Slide 2 | Roadmap

Guide Tell students that today moves from a null hypothesis to a test, an interval, and finally a comparison of groups.

Slide 3 | Last lesson: probability

Guide Retrieve one fact: $\mathbb{E}(Y \mid X = 1)$ is a conditional mean, not the unconditional mean. Use the sunny-day example for a 30-second calculation.

Slide 4 | Last lesson: sampling

Guide Ask why $\text{Var}(\bar{Y}) = \sigma^2/n$ matters for inference. Students should answer: it determines the scale of sampling noise.

Slide 5 | What is hypothesis testing?

Guide State H_0 and H_1 before looking at data. Example: test $H_0 : \mu = 100$ against $H_1 : \mu \neq 100$.

Slide 6 | What is hypothesis testing?

Guide Emphasize that the null is a reference model, not automatically the truth. Ask what a two-sided alternative allows.

Slide 7 | Why use hypothesis testing?

Guide Connect the procedure to a decision under uncertainty. Example: a drug trial where a sample improvement may be sampling variation.

Slide 8 | Key concepts

Guide Define test statistic, reference distribution, and p-value in that order. Ask which object is random under the null: the statistic, not the fixed parameter.

Slide 9 | Test statistic setup

Guide Put the observed mean and hypothesized mean on a number line. The distance must be scaled by its standard error.

Slide 10 | Computing the p-value

Guide Use the two-sided rule: count probability in both tails at least as far from zero as the observed statistic. Warn against reporting $P(H_0 \mid \text{data})$.

Slide 11 | Sample variance

Guide Calculate $s^2 = (n - 1)^{-1} \sum_i (y_i - \bar{y})^2$ and explain why the denominator is $n - 1$.

Slide 12 | Standard error

Guide For $n = 100$, $s = 20$, calculate $SE = 2$ on the board. Ask what happens to SE if n becomes 400.

Slide 13 | Type I and Type II errors

Guide Use a pregnancy-test analogy only to label false positive and false negative. Then return to econometrics: Type I rejects a true null; Type II fails to reject a false null.

Slide 14 | Errors in testing

Guide Explain the trade-off between a smaller significance level and power. Do not call failure to reject acceptance of the null.

Slide 15 | Confidence intervals

Guide Construct the interval as estimate plus and minus critical value times standard error. Ask what repeated-sampling coverage means.

Slide 16 | What are CIs?

Guide Use the interval to show that a two-sided 5 percent test and a 95 percent interval answer matching questions under the same reference distribution.

Slide 17 | Computing CIs

Guide Work $104 \pm 1.96(2) = [100.08, 107.92]$. Ask whether the interval contains 100 and what that implies for the corresponding test.

Slide 18 | Mean difference testing

Guide Define $\hat{\Delta} = \bar{Y}_1 - \bar{Y}_0$ and keep the group labels visible. Example: $70 - 65 = 5$.

Slide 19 | Mean difference testing

Guide Compute $SE(\hat{\Delta}) = \sqrt{s_1^2/n_1 + s_0^2/n_0}$. Ask why independent groups permit variances to add.

Slide 20 | Randomized control trials

Guide Explain why random assignment makes a difference interpretable as a causal effect under the design. It does not guarantee equal groups in one finite sample.

Slide 21 | California data: setup

Guide Name the outcome, treatment or regressor, unit, and sample before interpreting any coefficient.

Slide 22 | California data: descriptive comparison

Guide Ask students to compare average test scores by class size category without calling the gap causal.

Slide 23 | California data: uncertainty

Guide Identify the estimate, standard error, and confidence interval in the displayed output. Have students point to each rather than reading the table aloud.

Slide 24 | California data: test

Guide Choose one null, write it symbolically, and calculate the corresponding statistic. Keep the null visible while interpreting.

Slide 25 | California data: interpretation

Guide Separate statistical significance from economic importance. Ask whether the interval rules out effects that would matter for policy.

Slide 26 | California data: limitations

Guide List selection, omitted variables, and measurement concerns. Correct arithmetic does not by itself create identification.

Slide 27 | California data: conclusion

Guide Have students give a complete sentence with estimate, units, uncertainty, and causal qualification.

Slide 28 | Review

Guide Use the synthetic two-group example from this guide as a one-minute retrieval exercise.

Slide 29 | Material

Guide Tell students which chapter pages and problem set questions to use for practice; do not spend class time reading the bibliography.

Board example | A mean

Use with slides Lecture2v2 slides 5–11: What is Hypothesis Testing? through the t-statistic. Start the numerical mean test on slide 8, compute the standard error on slide 10, and finish the test statistic on slide 11.

Illustrative sample: $n = 100$, $\bar{y} = 104$, $s = 20$. Test $H_0 : \mu = 100$ against $H_1 : \mu \neq 100$.

$$SE(\bar{Y}) = 20/\sqrt{100} = 2, \quad t = (104 - 100)/2 = 2.$$

Using a large-sample normal approximation, the two-sided p -value is about .0455 and a 95% interval is

$$104 \pm 1.96(2) = [100.08, 107.92].$$

Reject at 5% but not at 1%. If observations are i.i.d. normal and variance is estimated, an exact t_{99} reference gives a slightly larger p -value (about .048). State which reference distribution is being used.

What to say A p -value is the probability, under the null and assumptions, of a statistic at least as extreme as the observed one. It is not the probability that the null is true.

If short Keep the mean example and two-group example. Use one California output table deeply; assign repetitive descriptive tables for review.

Side example | Comparing groups

Use with slides Lecture2v2 slides 15–19: Confidence Intervals, Mean Difference Testing, and Causal Inference in Randomized Control Trials. Use the two-group calculation on slides 17–18, then discuss the random-assignment caveat on slide 19.

Two independently sampled groups have means 70 and 65, SDs 10 and 12, and sample sizes 50 each:

$$\hat{\Delta} = 5, \quad SE(\hat{\Delta}) = \sqrt{10^2/50 + 12^2/50} = \sqrt{4.88} \simeq 2.209.$$

$$t \simeq 2.264, \quad 95\% \text{ large-sample CI: } 5 \pm 1.96(2.209) = [.67, 9.33].$$

Ask students what changes if these are the same individuals measured twice. The independent-groups formula is then inappropriate; the covariance between measurements matters.

Causal interpretation If groups are assigned randomly and the outcome comparison follows that assignment, a mean difference can estimate a treatment effect under the design's assumptions. If students choose their own school, the same arithmetic does not remove selection.

Type I/II prompt A Type I error rejects a true null. A Type II error fails to reject a false null. Holding sample size and effect fixed, demanding a smaller significance level generally reduces power.

Confidence-interval language The method covers the fixed population parameter in 95% of repeated samples under its assumptions. Avoid saying that 95% of observations lie in this interval.

Economic significance An enormous sample can make a tiny effect statistically significant. Ask whether the interval excludes effects small enough to be irrelevant to the decision.

Exit exercise and answers

If $n = 400$ with the same SD and the same observed gap of four units, $SE = 1$ and $t = 4$. The standard error halves, not quarters.

If a 95% interval includes zero, fail to reject a two-sided zero-effect null at the corresponding 5% level. This is not proof of zero effect.

Next preparation Read Chapter 4. Bring a scatterplot of test scores against class size to the next class and ask how to summarize its slope.

Source: Lecture2v2.tex. The numerical data above are synthetic teaching examples.